

PYTHON - SE - BACKEND

SOFTWARE ENGINEERING – Introduction to Programming

MODULE - 2

SESSION / ASSIGNMENT – 7

TASKS :

1. Use nested for loops to print a grid of emojis representing a 5x5 Instagram post feed, where each cell shows a 📷 symbol.



```
#include <stdio.h>

int main()
{
    int i, j;

    for(i = 1; i <= 5; i++)
    {
        for(j = 1; j <= 5; j++)
        {
            printf("📷 ");
        }
        printf("\n");
    }

    return 0;
}
```

OUTPUT:



2. Build a right-angled triangle pattern using nested loops, where each row displays increasing numbers starting from 1, similar to how a leaderboard on a gaming app shows rank numbers.



```
#include <stdio.h>

int main()
{
    int i, j;

    for(i = 1; i <= 5; i++)
    {
        for(j = 1; j <= i; j++)
        {
            printf("%d ", j);
        }
        printf("\n");
    }

    return 0;
}
```

OUTPUT:

```
1
1 2
1 2 3
1 2 3 4
1 2 3 4 5
```

3. Create a pattern that prints a pyramid of stars (*) with 6 rows, centered like the loading animation you see on BookMyShow when a page is loading
- Hint: Use spaces to align the stars in the center for each row.



```
#include <stdio.h>

int main()
{
    int i, j;

    for(i = 1; i <= 6; i++)
    {
        for(j = 1; j <= 6 - i; j++)
        {
            printf(" ");
        }

        for(j = 1; j <= (2 * i - 1); j++)
        {
            printf("*");
        }

        printf("\n");
    }

    return 0;
}
```

OUTPUT:

```
PS E:\OOPJ_Ne
Software_engi
*
***
*****
*****
*****
*****
```

4. Write code using nested loops to print a pattern of alternating 0s and 1s in a grid, like the checkered background seen in some Spotify playlist covers (e.g., for a 4x4 grid, alternate 0 and 1 in each cell).



```
#include <stdio.h>

int main()
{
    int i, j;

    for(i = 1; i <= 4; i++)
    {
        for(j = 1; j <= 4; j++)
        {
            if((i + j) % 2 == 0)
                printf("0 ");
            else
                printf("1 ");
        }

        printf("\n");
    }

    return 0;
}
```

OUTPUT:

```
PS E:\OOP\
Software e
0 1 0 1
1 0 1 0
0 1 0 1
1 0 1 0
```

5. Modify your pyramid pattern code to accept the number of rows as user input, so the user can set the height of the pyramid before printing.



```
#include <stdio.h>

int main()
{
    int rows, i, j;

    printf("Enter the number of rows: ");
    scanf("%d", &rows);

    for(i = 1; i <= rows; i++)
    {
        for(j = 1; j <= rows - i; j++)
        {
            printf(" ");
        }

        for(j = 1; j <= (2 * i - 1); j++)
        {
            printf("*");
        }

        printf("\n");
    }

    return 0;
}
```

OUTPUT:

```
Enter the number of rows: 7
  *
 ***
*****
*****
*****
*****
*****
```